

Chapter 1

Materials and methods

1.1 Materials and equipment

Table 1.1: Microfluidic chips, instrumentation, optical components, and reagents used in the study.

Item	Key specifications	Cat. No.	Src.
Microfluidic chips and consumables			
Droplet generation chip	Cruciform, flow-focusing type; Topas; nominal channel diameter 10–30 μm	Fluidic 947	MCS
Droplet generation chip	Cruciform, flow-focusing type; Topas; nominal channel diameter 30–60 μm	Fluidic 948	MCS
Measuring reaction chip	PMMA; thickness 100–800 μm ; volume 1.2–10.2 μL	Fluidic 621	MCS
Glass measuring chip	Borosilicate glass; depth 30 μm x 3.1 mm	Fluidic 1075	MCS
Additional equipment for microfluidics			
Mini-Luer M-type	TPE material	Fluidic 331	MCS
Mini-Luer F-type	PP material	Fluidic 631	MCS
Syringe adapters	Polypropylene; Luer interface	—	MCS
Syringe	1 mL HJT; PP	—	HSW
Sealing cover film	Type 631-0144	631-0144	VWR
Microfluidic instrumentation			
Injection pump	AL-1000; 12 V; 1000 mA; 0.001 $\mu\text{L}/\text{h}$ (0.5 μL syringe) to 3470 mL/h (140 mL syringe); 1%	AL-1000	WPI
Coil element	VISATON FC 6.8–10 mH; 31 mm x 40 mm; wire 0.8 mm; 0.75 Ω ; I_{sat} 5.4 A	FC 6.8–10 mH	RCH
Magnetic modulation hardware			
Power supply	KORAD CA3005DS; 0–30 V DC / 0–5 A	10 mV	RCH
Microcontroller board	Arduino Uno Rev. 3; ATmega328; 16 MHz	A000073	ARD
Air cooling module	SBC-303010; 7500 RPM; 12 V	SBC-303010	RCH
Peltier elements	B09-732 TECB13; 20 $\times$ 20 mm	-	
Baseboard	DELBO v44x55	-	RCH

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Item	Key specifications	Cat. No.	Src.
H-bridge motor driver	DRV 8424P	DRV 8424P	RCH
Gaussmeter	IDR 309A	IDR 309A	Integrity Design
3D printer	Creality Ender 3V3	Ender 3V3	RCH
Optical and imaging system			
Collimated LED	M470L2-C4 Blue; 470 nm; 1600 mA	M470L2-C4	THL
LED driver	LEDD1B; 1200 mA; 0–5 kHz	LEDD1B	THL
Excitation filter	FITC; $\varnothing$ 25 mm; 479/15 nm	—	THL
Emission filter	FITC; $\varnothing$ 25 mm; 535/20 nm	—	THL
Dichroic mirror	380–497 nm / 514.5–900 nm	—	THL
Objective lens	SN6/LCI Plan-Neofluar 25 $\times$ 0.8 Imm Corr DIC M27; CG = 0–0.17 mm; FWD = 0.21 mm at CG = 0.17 mm	—	ZEI
Tube lens	ACL12104	ACL12104	THL
Camera	a2A3840-45um BAS; Sony IMX334 CMOS sensor; sensor size 7.7 $\times$ 4.3 mm; 45 fps; mono8	a2A3840- 45um	BAS
Optical breadboard	Through-tapped 1/4"-20 (M6) aluminium	—	THL
Cage plates	30 mm cage plates CP6T, CP02F, CP32/M	-	THL
Cage system rods	30 and 60 mm ; $\varnothing$ 6 mm	—	THL
Optical rails	25 mm 225–600 mm, M6 T-nuts	—	THL
Chemical and biological reagents			
Fluorinated oil	Fluorochem FD-128; 100 g;	F567667	FCH
Surfactant	dSURF 2% 3M TM Novec TM 7500 oil		FLG
iLOV protein	pGEX-4T-1	pGEX-4T-1	GEH
FAD	95%; 50 mg	FAD282- 733H	ROT
HEWL	39 K units/mg protein; CAS 12650-88-3	L6876-5G	SIG
Citric acid	ACS reagent grade; 99.5%; 77-92-9	—	SIG
Tryptophan	99.5% by HPLC; dry; CAS 73-22-3	—	SIG

Abbreviations: MCS – Microfluidic Chip Shop, Jena, Germany; WPI – World Precision Instruments Germany GmbH, Friedberg, Germany; RCH – Reichelt Elektronik GmbH, Sande (Friesland), Germany; THL – Thorlabs GmbH, Bergkirchen, Germany; ZEI – Carl Zeiss Microscopy LLC, Germany; BAS – Basler AG, Germany; ARD – Arduino, Turin, Italy; FCH – Fluorochem Pvt, India; FLG – Fluigent, France; GEH – GE Healthcare GmbH, Germany; ROT – Carl Roth GmbH, Germany; SIG – Sigma-Aldrich, USA.